

DEEP DIVE

Maths for ML — Deep Dive

Calculus, Linear Algebra, Probability & Statistics — thoroughly

 8 weeks

 Intermediate

 Live+Recorded

 Prasant Mishra

ABOUT THIS COURSE

Eight weeks of rigorous mathematical foundations for machine learning. Goes significantly deeper than the weekend cohort — with proofs, derivations, and applied exercises every week.

Prerequisites: 12th grade maths. Maths for ML Weekend Cohort or equivalent.

WHAT YOU'LL LEARN

- 1 Derive and understand core ML algorithms mathematically
- 2 Read ML papers comfortably
- 3 Apply maths directly in code (NumPy)
- 4 Have a solid foundation for PhD-level AI work

CURRICULUM · 4 SESSIONS

SESSION WEEKS 1–2

Calculus for ML

Limits, continuity, differentiability

Partial derivatives & the gradient

Chain rule & backpropagation derivation

Multivariable optimisation & Lagrange multipliers

SESSION WEEKS 3–4

Linear Algebra

Vector spaces, basis & dimension

Matrix factorizations (LU, QR, SVD)

Eigendecomposition in depth

PCA derivation from scratch

SESSION WEEKS 5–6

Probability & Statistics

Probability theory — sigma algebras, random variables

Common distributions & their properties

Bayesian inference & MLE/MAP

Hypothesis testing & confidence intervals

SESSION WEEKS 7–8

Information Theory & Applications

Entropy, KL divergence, mutual information

These concepts in neural networks

Reading ML paper math sections

Capstone: derive a classic ML algorithm

WHAT'S INCLUDED

- ✓ 16 live sessions + all recordings
- ✓ Weekly problem sets with full solutions
- ✓ NumPy code notebooks
- ✓ Learner's Certificate

UPCOMING BATCHES



Ongoing · Sat & Sun · 10:00 AM – 12:00 PM IST



New batches start monthly



learner

₹12,999

Live+Recorded · 8 weeks

[Enrol Now →](#)